Mastering Agentic AI Certification: Week 5 – Project

Fine-Tune a Support Ticket Router

Submitted by: Punit Singh

Install Dependencies

```
Attempting uninstall: accelerate
  Found existing installation: accelerate 1.15.0
  Uninstalling accelerate-1.15.0:
    Successfully uninstalled accelerate-1.15.0
Attempting uninstall: peft
  Found existing installation: peft 0.21.2
  Uninstalling peft-0.21.2:
    Successfully uninstalled peft-0.21.2
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27/27 [llamafactory]
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts
hf-gradio 0.4.1 requires gradio-client<3.0,>=2.0, but you have gradio-client 1.14.0 which is incompatible.
gcsfs 2025.12.0 requires fsspec==2025.12.0, but you have fsspec 2025.3.0 which is incompatible.
google-genai 2.12.1 requires pydantic<3.0.0,>=2.12.5, but you have pydantic 2.12.3 which is incompatible.
python-fasthtml 0.14.18 requires starlette>=1.0.1, but you have starlette 0.52.1 which is incompatible.
google-adk 2.7.1 requires starlette<2,>=1.3.1, but you have starlette 0.52.1 which is incompatible.
Successfully installed accelerate-1.11.0 aiofiles-24.1.0 av-16.0.0 datasets-4.0.0 dill-0.3.8 ffmpy-1.0.0 fire-0.7.1 fsspec-2025.3.0 gradio-5.50.0 gradio-client-1.14.0 hf-tra
```

Prepare Support Ticket Dataset (INPUT REQUIRED)

```
Upload support_ticket.csv:
Choose Files support_tickets.csv
support_tickets.csv(text/csv) - 102054 bytes, last modified: 9/4/2026 - 100% done
Saving support_tickets.csv to support_tickets.csv
Loaded 585 rows
label
Support general      150
Fileservice          138
O365                  92
EOL                   58
Software              58
Active Directory      48
Computer-Services     41
Name: count, dtype: int64

Train: 468 rows | Val (held-out): 117 rows

Train label distribution:
label
Support general      120
Fileservice          110
O365                  74
Software              46
EOL                   46
Active Directory      39
Computer-Services     33
Name: count, dtype: int64
```

```
KeyboardInterrupt

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/usr/local/bin/llamafactory-cli", line 6, in <module>
    sys.exit(main())
             ~~~~^^
  File "/content/LLaMA-Factory/src/llamafactory/cli.py", line 24, in main
    launcher.launch()
    ~~~~~~~~~~~~~~~^^
  File "/content/LLaMA-Factory/src/llamafactory/launcher.py", line 167, in launch
    run_web_ui()
    ~~~~~~~~~~^^
  File "/content/LLaMA-Factory/src/llamafactory/webui/interface.py", line 97, in run_web_ui
    create_ui().queue().launch(share=gradio_share, server_name=server_name, inbrowser=True)
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.13/dist-packages/gradio/blocks.py", line 2950, in launch
    self.block_thread()
    ~~~~~~~~~~~~~~~~~^^
  File "/usr/local/lib/python3.13/dist-packages/gradio/blocks.py", line 3047, in block_thread
    self.server.close()
    ~~~~~~~~~~~~~~~~~^^
  File "/usr/local/lib/python3.13/dist-packages/gradio/http_server.py", line 69, in close
    self.thread.join(timeout=5)
    ~~~~~~~~~~~~~~~~^^^^^^^^^^^
  File "/usr/lib/python3.13/threading.py", line 1095, in join
    self._handle.join(timeout)
    ~~~~~~~~~~~~~~~~~^^^^^^^^^
KeyboardInterrupt
Killing tunnel 0.0.0.0:7860 <> https://2b65678a7775726fde.gradio.live
```

Review Training — Loss Curve (INPUT REQUIRED)

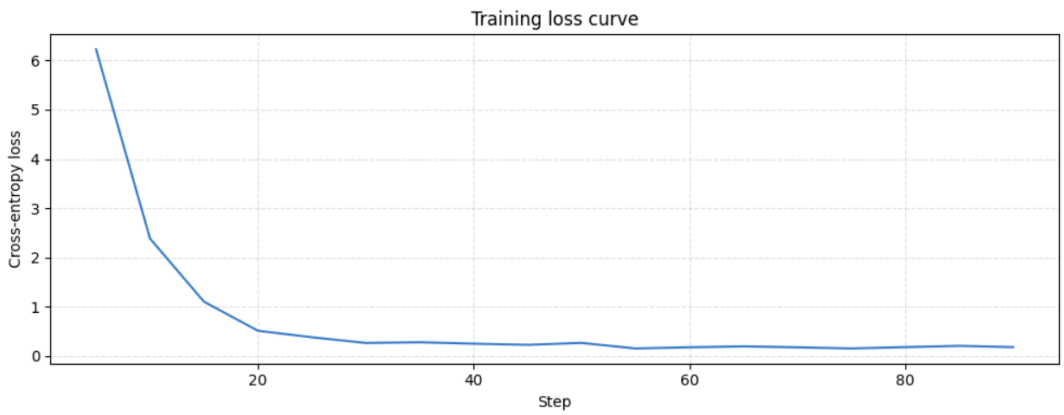
```python
import json, matplotlib.pyplot as plt
from pathlib import Path

ADAPTER_DIR = "/content/LLaMA-Factory/saves/Qwen3-1.7B-Base/lora/train_2026-11-08-14-40-01"  # <-- CHANGE THIS
MERGED_DIR      = "/content/qwen3_merged"
BASE_MODEL_NAME = "Qwen/Qwen3-1.7B-Base"

log_file = Path(ADAPTER_DIR) / "trainer_log.jsonl"
if not log_file.exists():
    raise FileNotFoundError(f"trainer_log.jsonl not found in {ADAPTER_DIR!r}")

records = [json.loads(l) for l in log_file.read_text().splitlines() if l.strip()]
steps   = [r["current_steps"] for r in records if r.get("loss") is not None]
losses  = [r["loss"]          for r in records if r.get("loss") is not None]

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(steps, losses, linewidth=1.5, color="#1565C0", alpha=0.85)
ax.set_xlabel("Step")
ax.set_ylabel("Cross-entropy loss")
ax.set_title("Training loss curve")
ax.grid(True, linestyle="--", alpha=0.4)
plt.tight_layout()
plt.savefig("/content/training_curve.png", dpi=120)
plt.show()
```




```
Starting loss : 6.2235
Final loss    : 0.1806
Total drop    : 6.0429

Loss is low — model has likely converged well.
```

Define classify() and Smoke Test the Merged Model

Ticket (truncated)	Expected	Predicted	Conf	
Please create a new user account for the new employee star	Active Directory	Support general	82.5%	--
I cannot access the shared network folder — I keep getting	Fileservice	Fileservice	64.3%	OK
Outlook keeps asking me to re-enter my password and Teams	O365	O365	79.1%	OK
We're ready to retire [SERVER] — please remove it from mon	EOL	EOL	95.8%	OK
Adobe Acrobat fails to install, error code 1603.	Software	Software	93.7%	OK

Evaluate the Fine-tuned Model on the Validation Split

```python
# --- 5.3  Evaluate fine-tuned model on val split ----------------------------
from tqdm.auto import tqdm

y_pred = []
for _, row in tqdm(df_val.iterrows(), total=len(df_val), desc="Fine-tuned inference"):
    pred, _ = classify(row["text"], compute_confidence=False)
    y_pred.append(pred)

print(f"Evaluated {len(y_pred):,} samples.")
```

```
Fine-tuned inference: 100% 117/117 [00:50<00:00,  4.66it/s]
Evaluated 117 samples.
```

```
                     precision    recall  f1-score   support

    Support general      0.500     0.222     0.308         9
        Fileservice      0.750     0.750     0.750         8
               O365      1.000     1.000     1.000        12
                EOL      0.920     0.821     0.868        28
           Software      0.650     0.722     0.684        18
     Active Directory      0.750     0.500     0.600        12
    Computer-Services      0.600     0.800     0.686        30

           accuracy                          0.735       117
          macro avg      0.739     0.688     0.699       117
       weighted avg      0.743     0.735     0.728       117

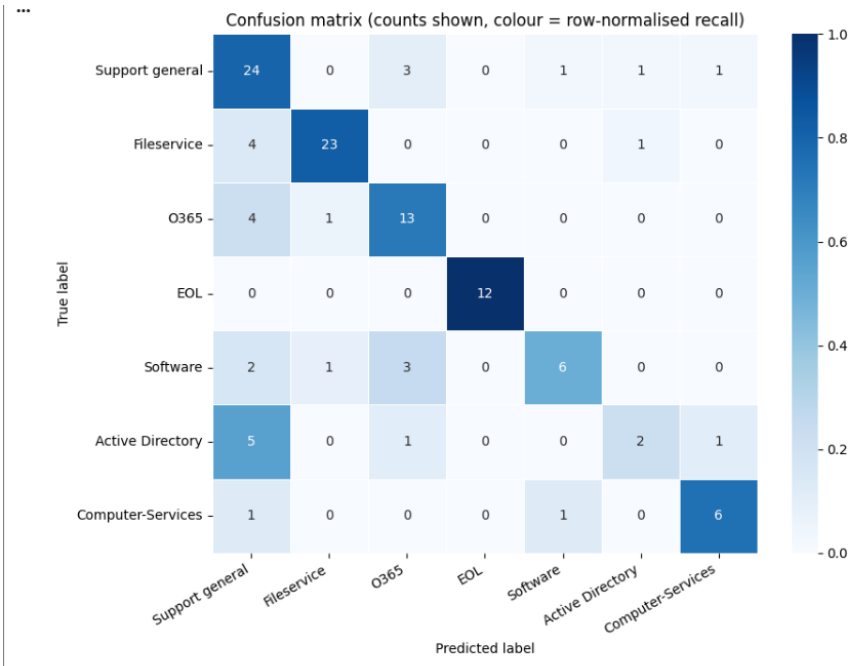

    === Ticket 0
    TRUE label: Fileservice
    PRED label: Fileservice  (conf 38.7%)

    === Ticket 10
    TRUE label: Active Directory
    PRED label: Support general  (conf 63.1%)

    === Ticket 20
    TRUE label: Computer-Services
    PRED label: Computer-Services  (conf 52.5%)

    === Ticket 30
    TRUE label: Fileservice
    PRED label: Fileservice  (conf 96.1%)
```

Confusion matrix




```
Active Directory recall : 22.2%
Fileservice recall      : 82.1%
```

Baseline vs Fine-tuned Comparison

